

```
In [1]: import pandas as pd
import numpy as np
import xgboost as xgb
from sklearn.metrics import r2_score, mean_absolute_error, root_mean_squared_error
from sklearn.model_selection import train_test_split
```

```
In [3]: df_amz = pd.read_excel("Complete_Amazon_dataset.xlsx")
df_amz
```

Out[3]:

	product-name	Sum of Net_Price	Sum of absolute-quantity-gms	Total Orders	total_revenue	Running_total	cummulative_
0	FZYEZY Natural Freeze Dried Mango Fruit for Ki...	115	20	1	115	115	0.0000
1	FzyEzy Freeze-Dried Jain Kadhi Khichdi -	150	48	1	150	265	0.0000
2	Hamps Heamwalt Protein Healthy Powder Mix whey...	185	200	1	185	450	0.0000
3	FZYEZY Natural Freeze Dried Custard Apple Powd...	195	50	1	195	645	0.0000
4	FZYEZY Natural Freeze Dried Black Jamun Cubes ...	200	40	1	200	845	0.0000
...	...	...	...	...	...	...	...
147	FzyEzy Healthy Premium Freeze-Dried Strawberry...	18062	2484	108	1950696	21835522	0.6454
148	FzyEzy Healthy Premium Freeze-Dried Strawberry...	30456	15652	74	2253744	24089266	0.7120
149	FZYEZY Natural	25545	6550	110	2809950	26899216	0.7951

	product-name	Sum of Net_Price	Sum of absolute- quantity- gms	Total Orders	total_revenue	Running_total	cummulative_
	Freeze Dried Black Jamon Powder...						
150	FzyEzy Healthy Premium Freeze- dried Prickly Pe...	31578	6200	101	3189378	30088594	0.88939
151	FzyEzy Premium Organic Barley Grass Powder	31983	30418	117	3742011	33830605	1.00000

152 rows × 7 columns

```
In [5]: df_qcom = pd.read_excel("Q_Commerce_Data_rating_Count_rework.xlsx")
df_qcom
```

Out[5]:

	Brand	Quantity (in Grams)	MRP	Price	Price / 100 Grams	Discount (%)	Rating	Rating Count
0	Happilo	150	649	549	366.00	15	4.5	846
1	Happilo	200	465	249	124.50	46	4.4	1300
2	Happilo	400	825	665	166.25	31	4.4	4000
3	Happilo	250	125	59	23.60	52	4.2	1900
4	Happilo	500	450	315	63.00	30	4.5	1300
...	...	...	...	...	...	...	...	...
70	Healthy Master	250	394	301	120.40	24	4.4	121
71	Healthy Master	100	141	109	109.00	23	4.2	192
72	Healthy Master	100	141	109	109.00	23	4.6	242
73	Healthy Master	30	33	30	100.00	9	4.5	202
74	Healthy Master	100	141	109	109.00	23	4.7	195

75 rows × 8 columns

```
In [7]: df_qcom.columns = df_qcom.columns.str.strip()
df_qcom.columns
```

```
Out[7]: Index(['Brand', 'Quantity (in Grams)', 'MRP', 'Price', 'Price / 100 Grams',
              'Discount (%)', 'Rating', 'Rating Count'],
              dtype='object')
```

```
In [14]: df_amz.columns
df_amz.columns = df_amz.columns.str.strip()
df_amz.columns
df_amz.rename(columns={'Sum of absolute-quantity-gms':'Quantity (in Grams)', 'Sum of
df_amz
```

Out[14]:

	product-name	Price	Quantity (in Grams)	Total Orders	total_revenue	Running_total	cummulative_%
0	FZYEZY Natural Freeze Dried Mango Fruit for Ki...	115	20	1	115	115	0.000003
1	FzyEzy Freeze- Dried Jain Kadhi Khichdi -	150	48	1	150	265	0.000008
2	Hamps Heamwalt Protein Healthy Powder Mix whey...	185	200	1	185	450	0.000013
3	FZYEZY Natural Freeze Dried Custard Apple Powd...	195	50	1	195	645	0.000019
4	FZYEZY Natural Freeze Dried Black Jamun Cubes ...	200	40	1	200	845	0.000025
...	...	...	...	...	...	...	...
147	FzyEzy Healthy Premium Freeze- Dried Strawberry...	18062	2484	108	1950696	21835522	0.645437
148	FzyEzy Healthy Premium Freeze- Dried Strawberry...	30456	15652	74	2253744	24089266	0.712055
149	FZYEZY Natural Freeze	25545	6550	110	2809950	26899216	0.795115

	product-name	Price	Quantity (in Grams)	Total Orders	total_revenue	Running_total	cummulative_%
	Dried Black Jamun Powder...						
150	FzyEzy Healthy Premium Freeze-dried Prickly Pe...	31578	6200	101	3189378	30088594	0.889390
151	FzyEzy Premium Organic Barley Grass Powder	31983	30418	117	3742011	33830605	1.000000

152 rows × 7 columns

```
In [16]: features = [ 'Quantity (in Grams)', 'Price' ]
         target = 'Rating Count'
```

```
In [18]: df_qcom_features = df_qcom[features].copy()
         df_qcom_target = df_qcom[target].copy()
```

```
In [20]: df_qcom_features
         df_qcom_target
```

```
Out[20]: 0      846
         1     1300
         2     4000
         3     1900
         4     1300
         ...
        70      121
        71      192
        72      242
        73      202
        74      195
         Name: Rating Count, Length: 75, dtype: int64
```

```
In [23]: df_qcom_features.fillna(df_qcom_features.median,inplace=True)
         df_qcom_target.fillna(df_qcom_target.median,inplace=True)
```

```
In [25]: df_qcom_features_train, df_qcom_features_test , df_qcom_target_train , df_qcom_targ
         model = xgb.XGBRegressor(
             objective= "reg:squarederror",
             n_estimator= 100,
             learning_rate= 0.1,
             max_depth= 5,
             random_state= 42 )
```

```
model.fit(df_qcom_features_train,df_qcom_target_train)
predicted_values = model.predict(df_qcom_features_test)
```

C:\Users\Ritwik Singh\AppData\Roaming\Python\Python313\site-packages\xgboost\training.py:200: UserWarning: [19:11:31] WARNING: C:\actions-runner_work\xgboost\xgboost\src\learner.cc:782:
Parameters: { "n_estimator" } are not used.

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
In [28]: rmse = root_mean_squared_error(df_qcom_target_test,predicted_values)
mae = mean_absolute_error(df_qcom_target_test,predicted_values)
r2 = r2_score(df_qcom_target_test,predicted_values)*100
```

```
In [30]: print(f"Root Mean Squared Error = {rmse:.2f}")
print(f"mean_absolute_error = {mae:.2f}")
print(f"R Squared Error = {r2:.2f} %")
```

Root Mean Squared Error = 2594.80
mean_absolute_error = 1696.09
R Squared Error = 82.04 %

```
In [32]: df_amz_features = df_amz[features].copy()
df_amz_features
```

```
Out[32]:
```

	Quantity (in Grams)	Price
0	20	115
1	48	150
2	200	185
3	50	195
4	40	200
...	...	...
147	2484	18062
148	15652	30456
149	6550	25545
150	6200	31578
151	30418	31983

152 rows × 2 columns

```
In [33]: df_amz_features.fillna(df_amz_features.median, inplace= True)
```

```
In [35]: predicted_values_amz = model.predict(df_amz_features)
predicted_values_amz = np.maximum(0,predicted_values_amz)
predicted_values_amz = np.round(predicted_values_amz).astype(int)
```

```
In [38]: df_amz["Projected_ratings"] = predicted_values_amz  
df_amz.to_excel("amazon_predicted_ratings_rework.xlsx")
```